



DIGITAL TECHNOLOGY

Academic Year 2025-26

Prof. Barbara PERNICI

Exam 23/7/2026

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- The exam is closed-book.
- The use of cellular phones or any other electronic devices during the exam is forbidden.

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained; lists of bullet items are insufficient to answer a question.

Please write the answer for the Python exercise on a separate piece of paper.

Question 1 [11 points]

Illustrate the differences between databases and data warehouses and describe the types of operations that can be performed in each of them.

Question 2 [11 points]

Explain how the iterative and collaborative approach of agile project management can be used to address the limitations of the linear and sequential approach of traditional project management, and provide examples of how agile project management can help to improve project outcomes in uncertain and complex environments.

[please turn over]

Question 3 [11 points]

1. Consider the following fragment of Python code:

```
fruits = ["apple", "banana", "apple", "cherry", "banana"]
counts = {}

for f in fruits:
    if f in counts:
        counts[f] += 1
    else:
        counts[f] = 1
```

Modify the code so that the information about the fruit positions (indices) are also stored.

2. After executing this Python code:

```
import pandas as pd

data = {
    "employee": ["Anna", "Marco", "Anna", "Luca", "Marco"],
    "hours": [40, 35, 42, 38, 30],
    "dept": ["Sales", "IT", "Sales", "IT", "Sales"]
}
df = pd.DataFrame(data)

def compute_bonus(h):
    if h > 40:
        return h * 2
    else:
        return h * 1.5

df["bonus"] = df["hours"].apply(compute_bonus)
summary = df.groupby("dept").bonus.sum()
```

What is the content of the variable summary?

3. Consider the following code:

```
import pandas as pd

enrollments = pd.DataFrame({
    'student': ['Anna', 'Marco', 'Anna', 'Luca', 'Marco'],
    'course': ['Math', 'Physics', 'Physics', 'Math', 'Chemistry']
})

teachers = pd.DataFrame({
    'course': ['Math', 'Physics', 'Biology'],
    'teacher': ['Dr. Scott', 'Dr. Schrute', 'Dr. Halpert']
})

join_type = ...
merged = enrollments.merge(teachers, on='course', how=join_type)
```

The join type has not been defined. Which types of join would make Dr. Halpert appear in the merged results?